

Teaching Exam

General**Science**

Force & laws of motion

Pressure

Lecture No.- 1



By- Sarika Ma'am

Recap of Previous Lecture

Topic

Units & Measurement



Topics to be Covered



Topic

Motion

Force

✓ **Types of Forces**

✓ **Pressure**

Laws of Motion



Topic : SHM



TEACHING
WALLAH

Motion Related terms

Distance (दूरी)

⇒ Actual path

⇒ वास्तविक तय किया गया रास्ता

⇒ Unit = meter

⇒ Scalar

⇒ Never be zero.

Displacement (विस्थापन)

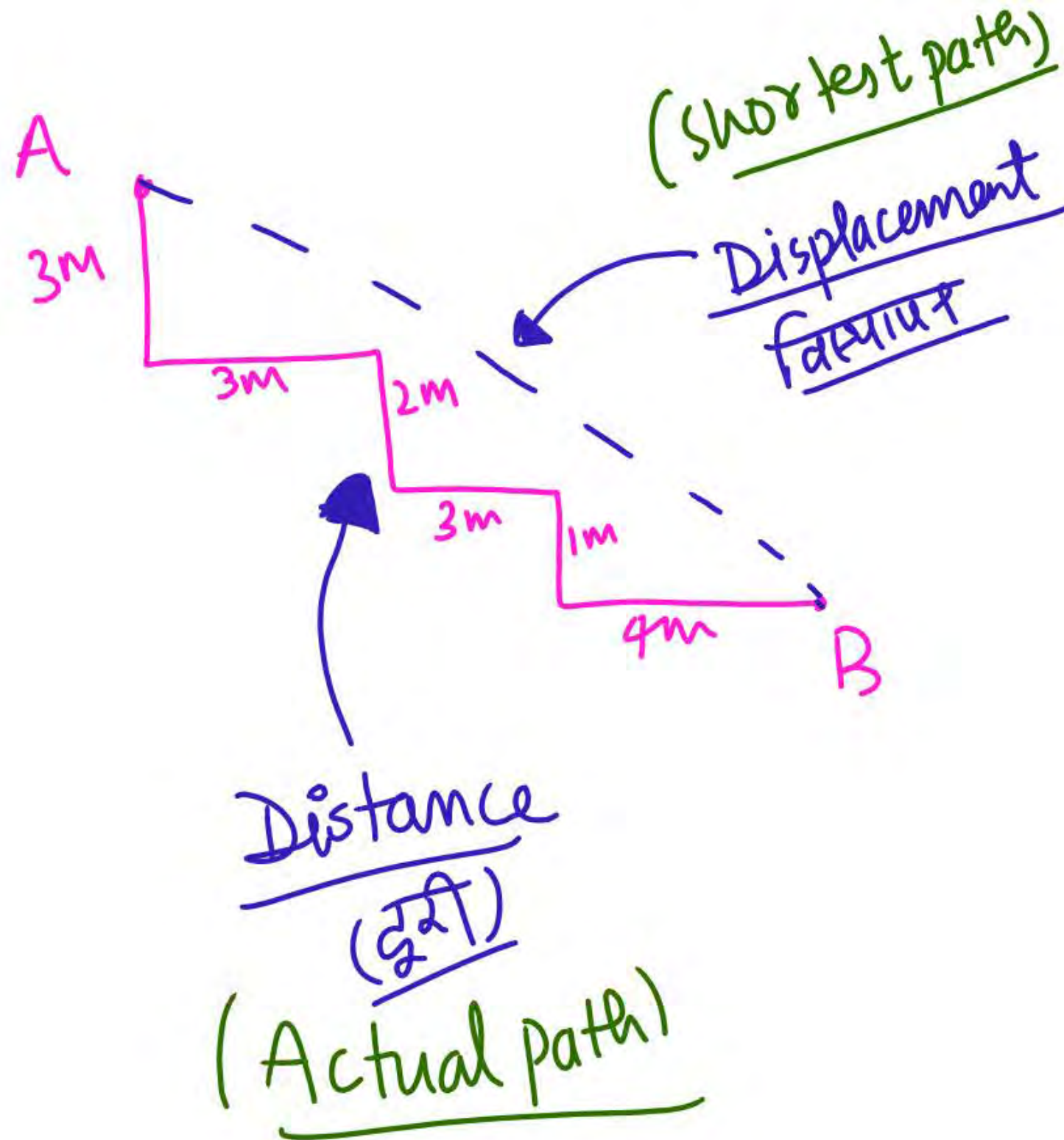
⇒ Shortest path

⇒ displacement = final position -
initial position

⇒ Unit = meter (m)

⇒ Vector (सदिश)

⇒ It can be zero.



Ques

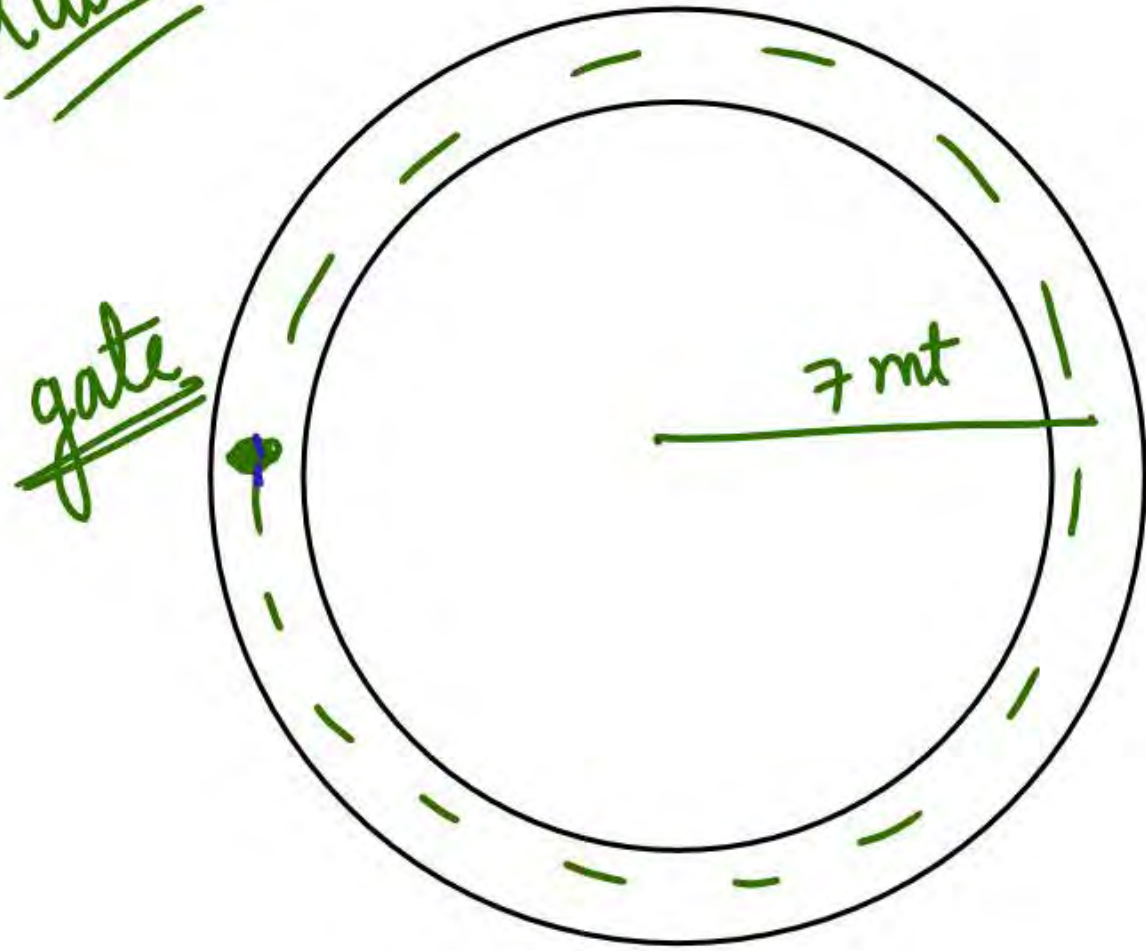


A से B से A तक-

$$\text{Total distance} = 5 + 5 = \underline{10 \text{ m}}$$

$$\text{Total displacement} = \underline{0} \quad \underline{\underline{\text{Ans}}}$$

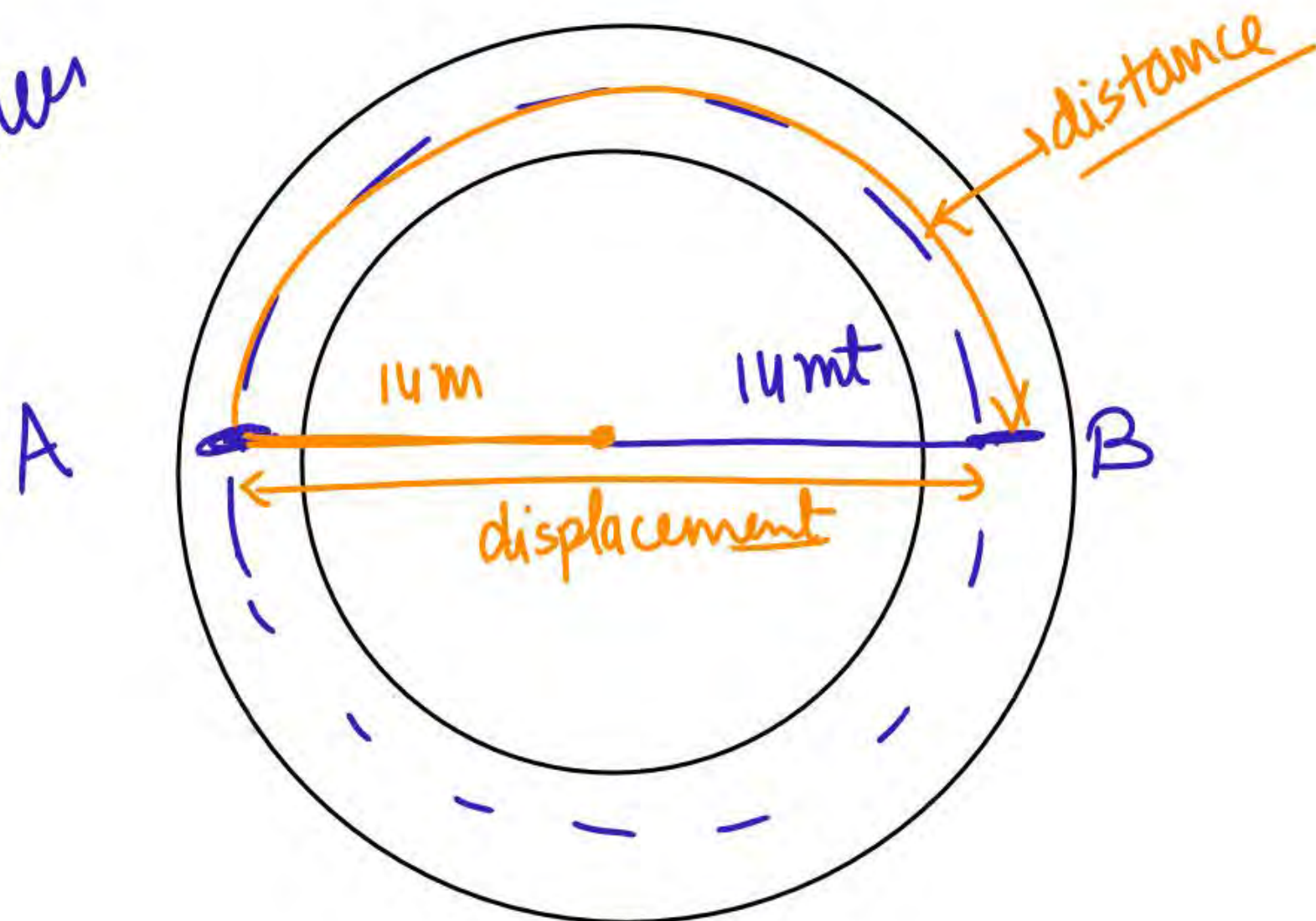
Ans



1 चक्कर में तय की गयी दूरी = $2\pi r$
 (distance) = $2 \times \frac{22}{7} \times 7$
 = 44 m

विस्थापन
 (displacement) = 0 Ans

Ques



A से B तक

Total distance =

$$\frac{2\pi r}{2} = \frac{22 \times 14^2}{7}$$

$$= \underline{44mt}$$

Total displacement =

$$14 + 14 = \underline{28mt}$$

Speed (चाल)

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

⇒ distance per unit time.
एकानक समय में चली गयी दूरी को चाल कहते हैं।

$$\Rightarrow \text{Unit} = \text{m/sec}$$

⇒ Scalar (अदिश)

Velocity (वेग)

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}}$$

$$\boxed{\text{वेग} = \frac{\text{विस्थापन}}{\text{समय}}}$$

→ एकांक समय में हुए विस्थापन को वेग कहते हैं।

$$\Rightarrow \text{Unit} = \text{m/sec}$$

⇒ Vector (अदिश)

Acceleration

त्वरण

Rate of change of velocity.
वेग परिवर्तन की दर को त्वरण कहते हैं।

$$\text{Acceleration} = \frac{v - u}{t}$$

unit

$$\text{m/sec}^2$$

Vector

वेग में परिवर्तन-

20m/sec → 80m/sec

जड़ रहा है।

$$a = +ve$$

80m/sec → 10m/sec

वेग परिवर्तन-

घट रहा है।

$$\Rightarrow a = -ve$$

Retardation

मंदन

Wrt $\boxed{\text{object} \rightarrow \text{Rest}}$ $\Rightarrow$ $\boxed{u=0}$ ✓

एक समान त्वरण के लिए गैलीलियो के समीकरण-

1) $v = u + at$

2) $s = ut + \frac{1}{2}at^2$

3) $v^2 = u^2 + 2as$

* u = initial velocity
प्रारम्भिक वेग

* v = final velocity
अंतिम वेग

* s = दूरी

* t = समय

* a = त्वरण

Graph-

1) displacement - time Graph

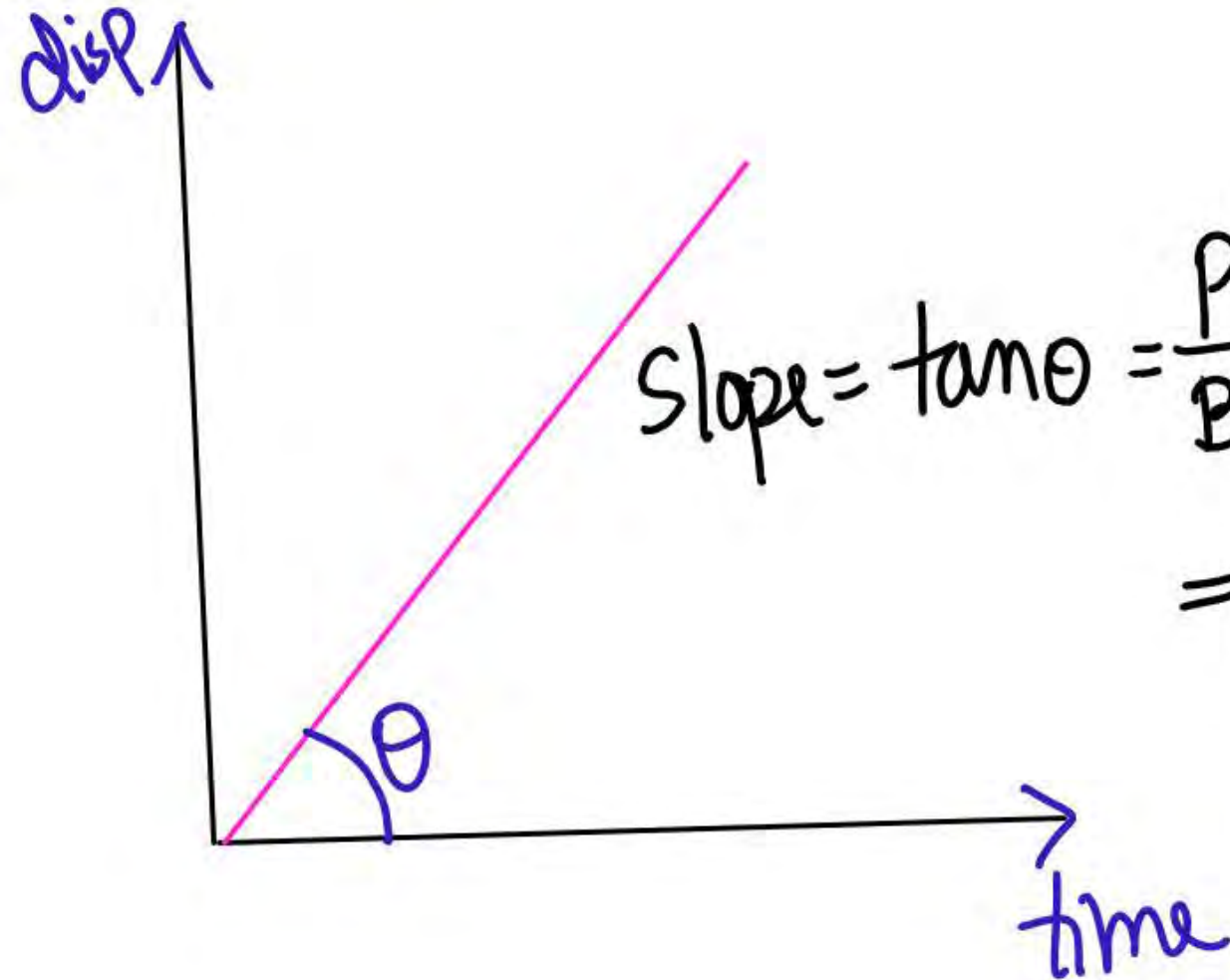
विस्थापन - समय ग्राफ

2) Velocity - time Graph

वेग - समय ग्राफ

Slope

Displacement-time Graph-

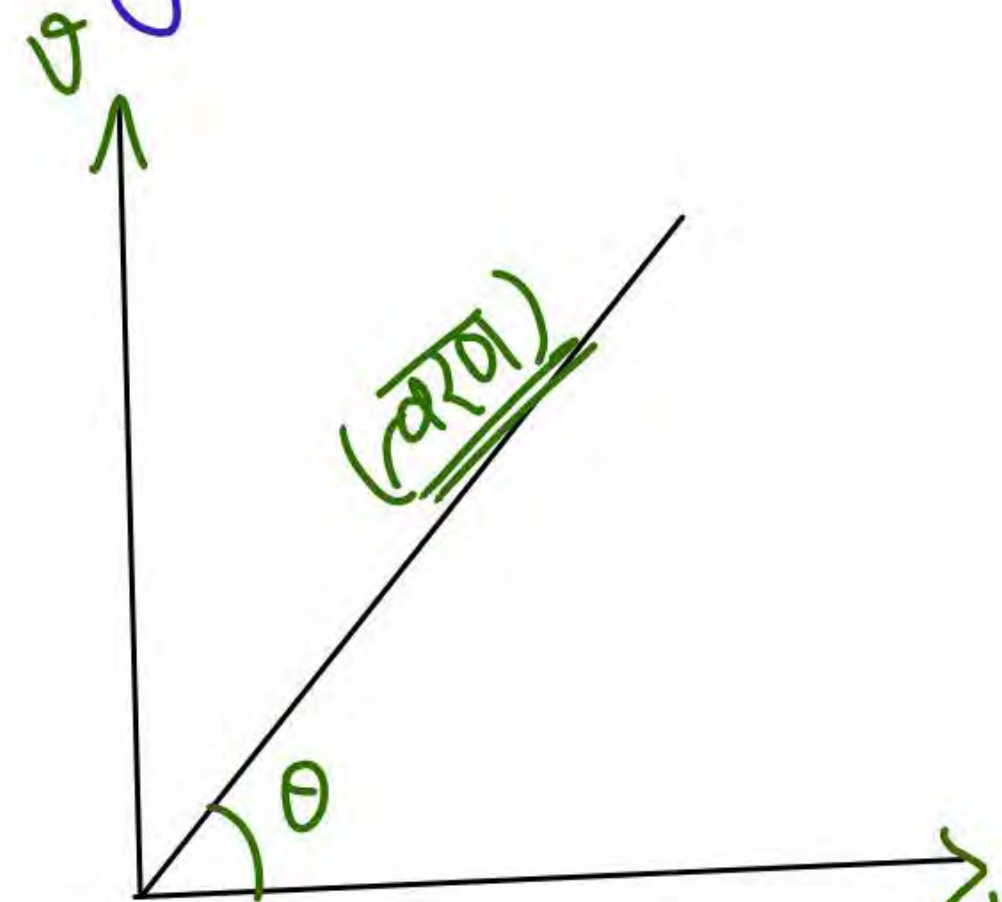


$$\text{Slope} = \tan \theta = \frac{P}{B} = \frac{\text{displacement}}{\text{time}}$$

$$= \text{Velocity}$$

वेग

Velocity-time Graph-

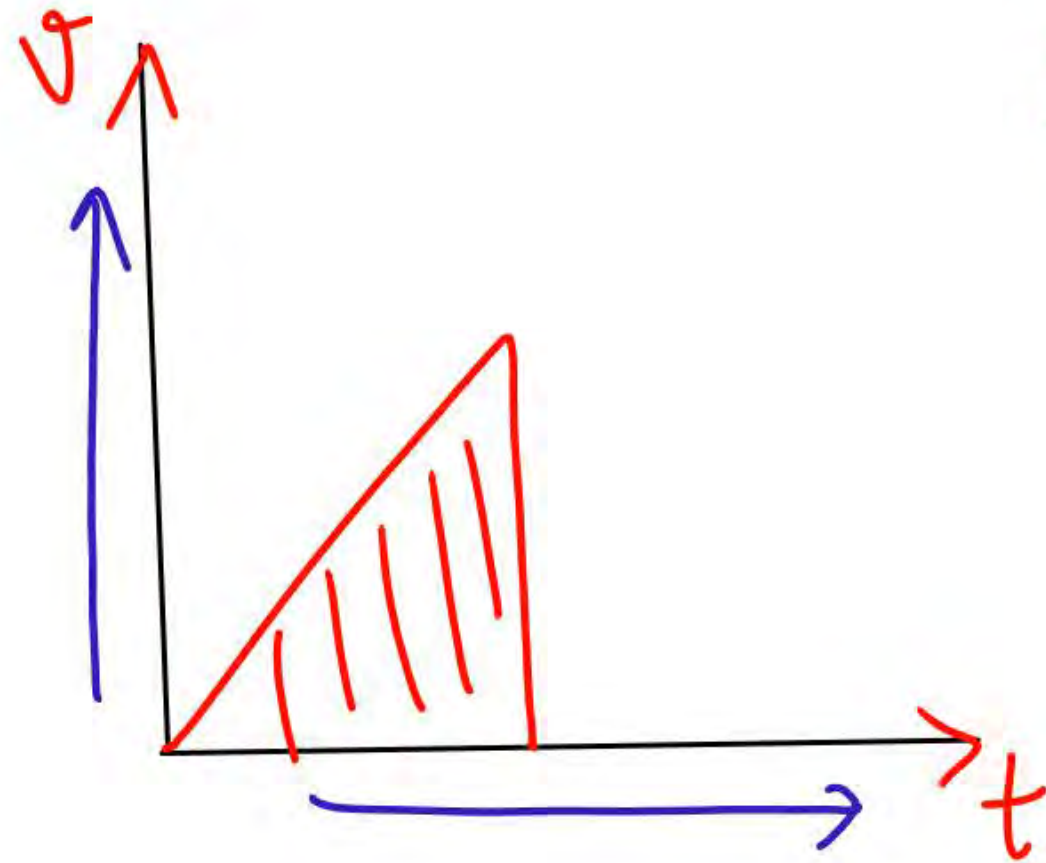


$$\text{Slope} = \tan \theta = \frac{P}{B} = \frac{\text{Velocity}}{\text{time}}$$

$$= \text{Acceleration (त्वरण)}$$

Area

Velocity-time Graph -



$$\text{Area} = \left(\frac{1}{2}\right) \text{base} \times \text{height}$$

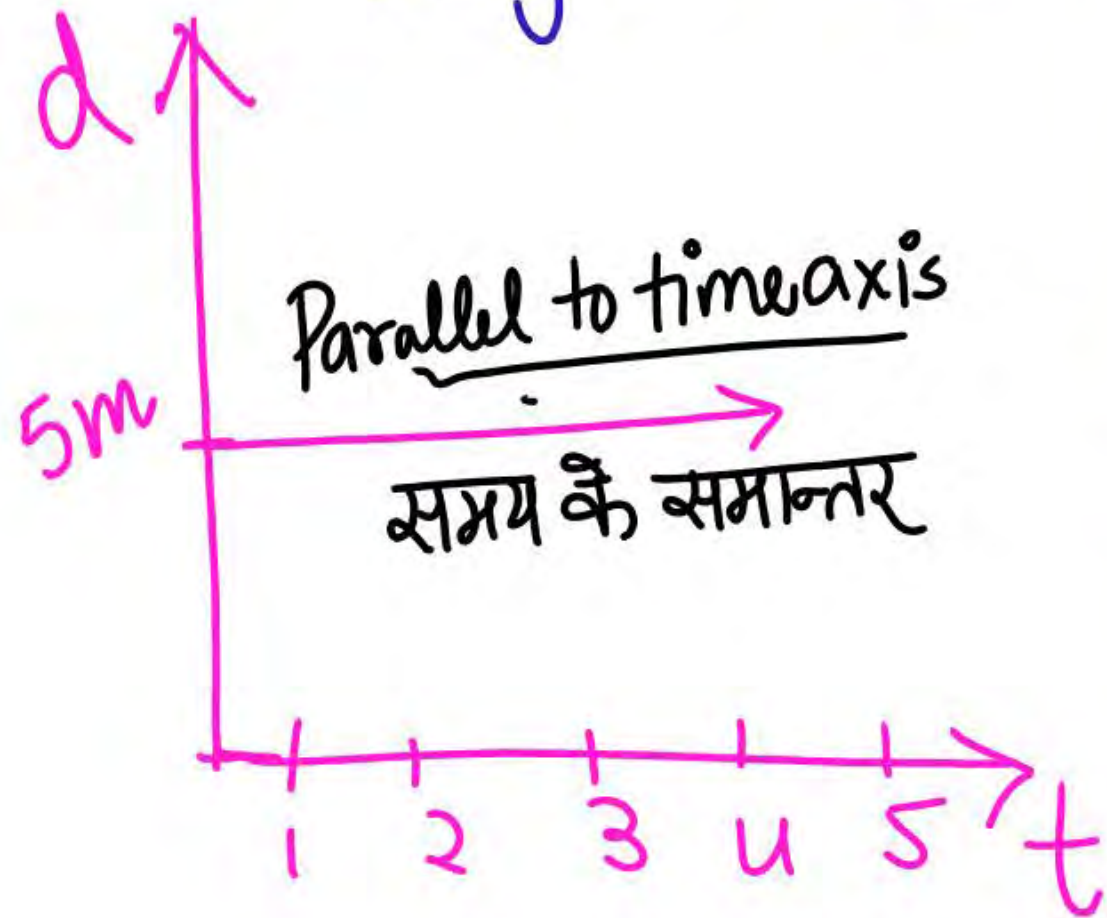
$$= \text{time} \times \text{velocity}$$

$$= \frac{\text{displacement}}{\text{time}}$$

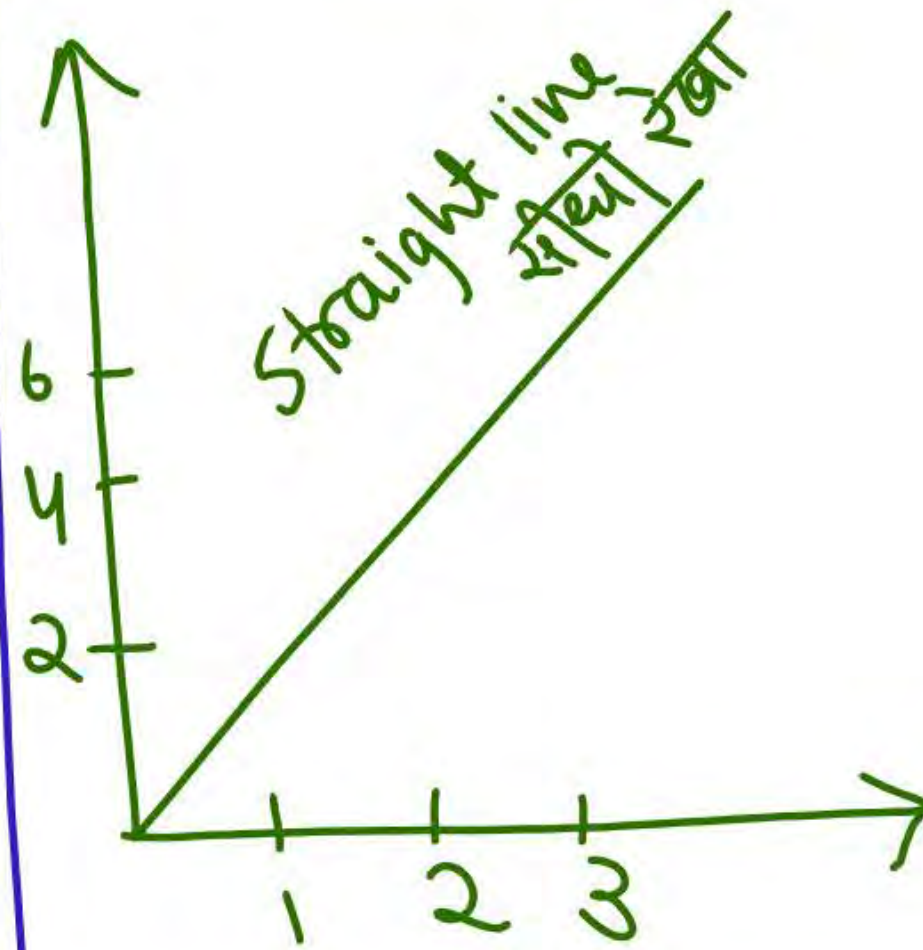
Graph-

Displacement-time Graph-

Object - Rest

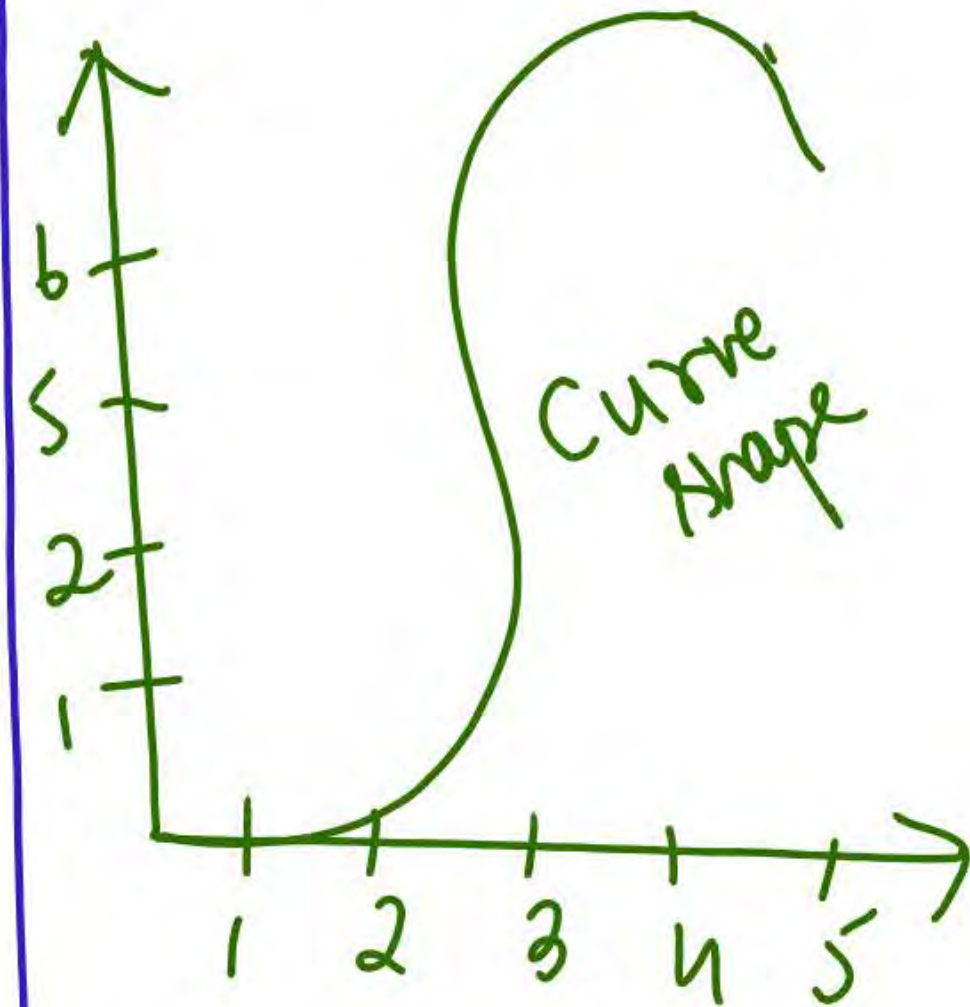


Uniform Motion



वेग स्थिर
Velocity const

Non-uniform



वेग बदल
velocity change



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Motion

Force, Laws of Motion, Pressure

THANK YOU